1 Real CHIME data

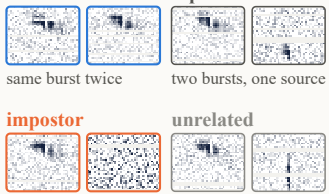


981 bursts · 83 repeaters

Catalog 2, 400 to 800 MHz

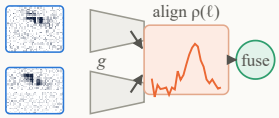
→ 950 noise-free templates
→ 856 real noise backgrounds
bursts split 60/20/20, by source

2 Pair simulator



+ nuisances per member: time shift, ΔDM, scattering, scintillation; frozen, hashed sets

3 Siamese correlation net



1. shared encoder, 2-sample resolution
2. soft alignment over ±24 samples
3. fuse $|g_A - g_B|$ and $g_A \odot g_B$
4. 4-class head → score s(A, B)

259k parameters

4 Validation, no real positives

- Pre-registered test — all 80 sources, run once — **+0.23 recall**
- Real cross-source pairs — vs per-pair twins — **FPR ≈ 1%**
- Real-burst copies — vs template copies — **ratio 0.97**
- Second-machine rerun — data, sets, training — **7/7 hashes**

5 Deployment: where can the detector work?

Timing test on burst lists (false alarms checked on 166,000 scrambles) and survey-geometry simulation → copies are recorded for only ~2% of lenses at a transit telescope; an all-sky monitor watching the same sources would find ~50× more lensed repeaters.